

Multi-Page Visually Rich Document Understanding

A survey of system design and an empirical attribution of failure

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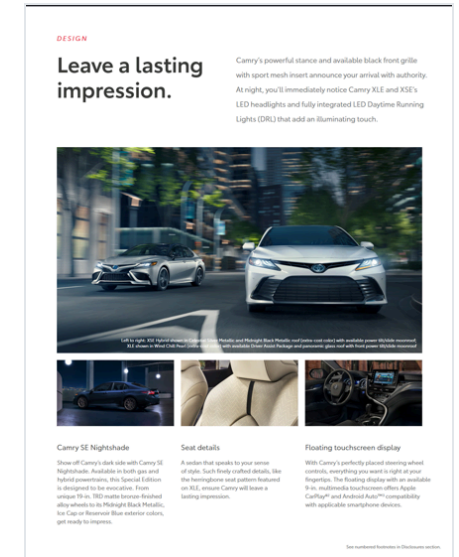
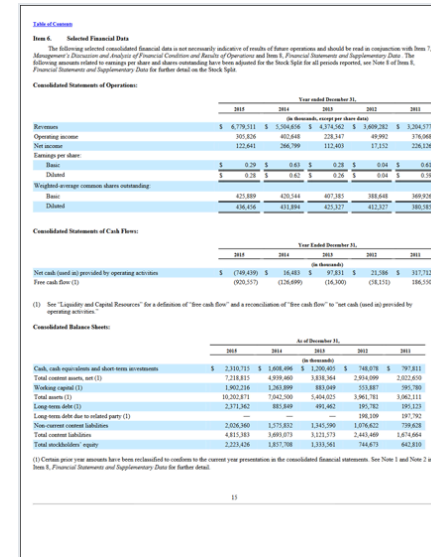
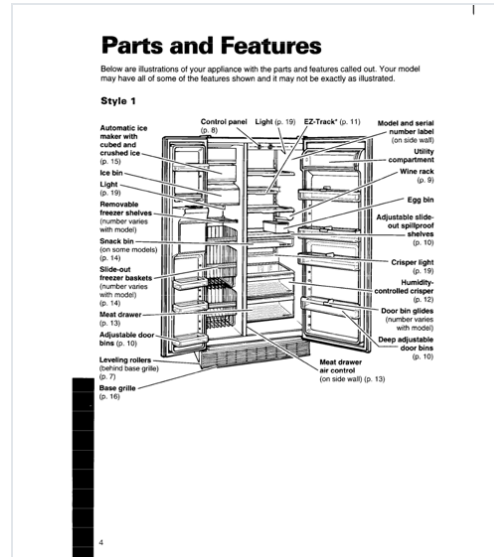
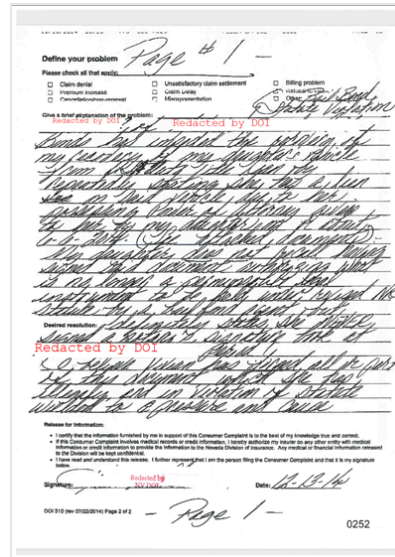
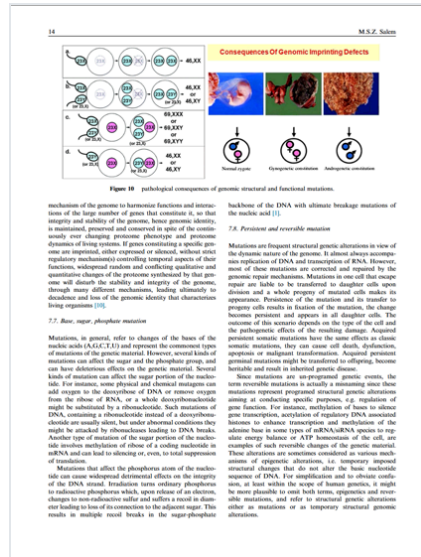


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- Introduction
- Survey of MP-VRDU
- Empirical Attribution Study
- Proof of Concept: A Simple Method
- Conclusion

Multi-Page Visually Rich Documents

Meaning is carried jointly by **text, vision, and layout** rather than by text alone. **Multi-page** adds a further layer: meaning also runs between pages, as when a claim points to a figure many pages away.

Research paper
born-digital

Signed form
scanned, handwritten

Technical manual
layout-dependent

Financial report
dense tables

Product brochure
photo-led

- **Text.** Small, typeset or handwritten, and legible only at far higher input resolution than a typical vision-language task requires.
- **Vision.** Charts, figures, photographs, and handwriting carry meaning that no text extraction recovers.
- **Layout.** Position is semantic, and reading order must be recovered rather than assumed.

What is Document Understanding?

Recover **what is on each page** and **how the whole document is organised**, then answer questions over it.

LAYOUT ANALYSIS

3
FEW RESEARCH CENTER

Latinos Increasingly Confident in Personal Finances, See Better Economic Times Ahead

Yet many economic indicators show few gains for the community since the Great Recession

Latinos have become considerably more upbeat about their personal finances and optimistic about their financial future since the Great Recession, according to newly released results from a national survey of Latino adults. The survey also shows that Latinos have pulled even with the general U.S. population in their views of their personal finances and continue to outpace them on optimism about the future. However, community economic indicators show limited progress since the Great Recession.

Four-in-ten Latinos say their personal finances are in "excellent" or "good" shape, a 17 percentage point increase since 2008, when only 23% made a similarly positive assessment of their finances (the Great Recession began in December 2007).¹ By contrast, the share of all Americans who have a similarly positive view of their finances remained essentially flat during the same seven-year period (41% in 2008 vs. 43% in 2015).

The survey also shows that Hispanics are more upbeat in their financial expectations for the upcoming year than they were in 2008. About eight-in-ten Hispanic adults (81%) say they expect their family's financial situation to improve "a lot" or "some" in the next year, up from 67% who said the same in 2008. By comparison, the U.S. public is not as upbeat –

Hispanics more confident in personal finances than in 2008, see better economic times ahead

% who say their personal financial situation is in excellent/good shape

% who expect their family's financial situation to improve "a lot" or "some" in the next year

Year	Hispanic	General public
2004	31	41
2008	23	41
2015	43	43

Year	Hispanic	General public
2004	70	67
2008	67	67
2015	81	64

Note: Shaded area marks the Great Recession, which ran from December 2007 to June 2009.
Source: National Survey of Latinos conducted Oct. 23-Nov. 30, 2015; surveys conducted Dec. 8-13, 2010; Oct. 24-28, 2012; Sept. 12-16, 2013; Dec. 7-11, 2011; Dec. 3-8, 2008; Oct. 19-22, 2006; Aug. 5-10, 2004.
¹Latinos Increasingly Confident in Personal Finances, See Better Economic Times Ahead

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The Great Recession lasted from December 2007 to June 2009, according to economists at the National Bureau of Economic Research.

www.fewresearch.org

Title Section heading Text Figure / chart Table Caption / note Header / footer

DOCUMENT OUTLINE

Research paper, 24 pages

5.2 Conscious Incompetence

Paragraphs

Figure 4

Caption

5.3 Retrieval Analysis

Paragraphs

Figure 5

Caption

5.4 Human Evaluation

Paragraphs

Table 6

Caption

LLM Assistants

Provide a demonstration of how you would use tool calls to answer a difficult question over this long visually rich document.

Designing a sample tool-based walkthrough for a cross-page question.

I'll demonstrate on a question that needs a figure and a table: "How often does the final pipeline stop at its earliest allowed point, and how does its page usage compare with the number of gold evidence pages?" First I'll map printed page numbers to PDF pages, then read the relevant pages as images.

Read a file, updated tasks ▾

Added task **Locate pages for Figure 4.2 and Table 4.2** ▾

Request
Description
Map printed page numbers to PDF pages using the contents list and the offset seen on page 27

Added task **Read figure and table pages visually** ▾

Request
Description
Read PDF pages 60-61 and extract the bar heights and token statistics

Added task **Verify and answer** ▾

Request
Description
Cross-check figure values against surrounding text, then answer

Completed task **Locate pages for Figure 4.2 and Table 4.2**

Started task **Read figure and table pages visually**

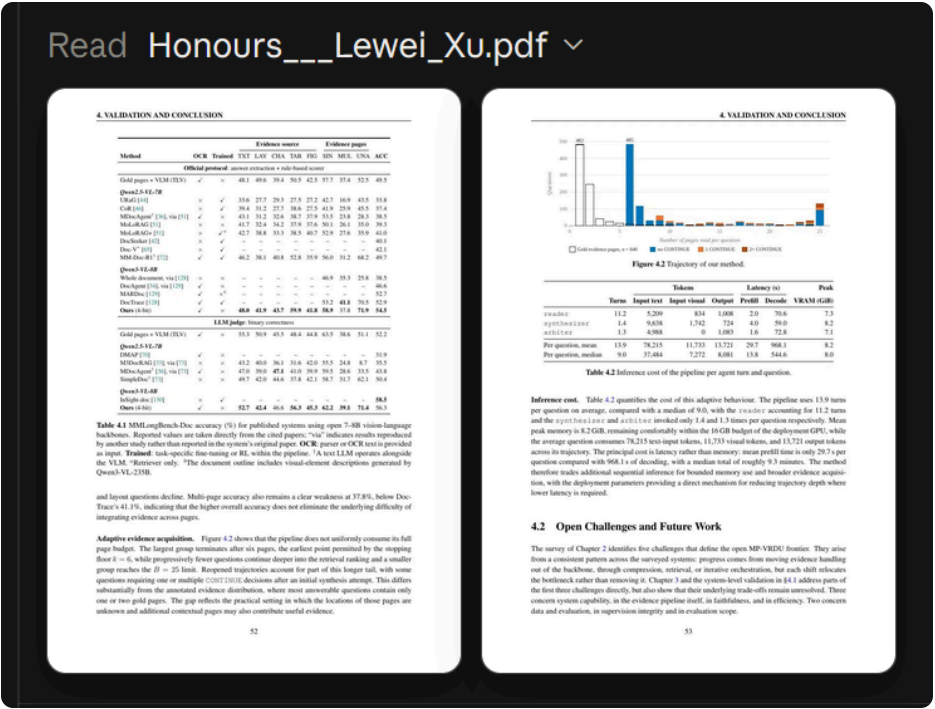
Read **Honours___Lewei_Xu.pdf** ▸

Verifying the data points plotted across the chart.

Completed task **Read figure and table pages visually**

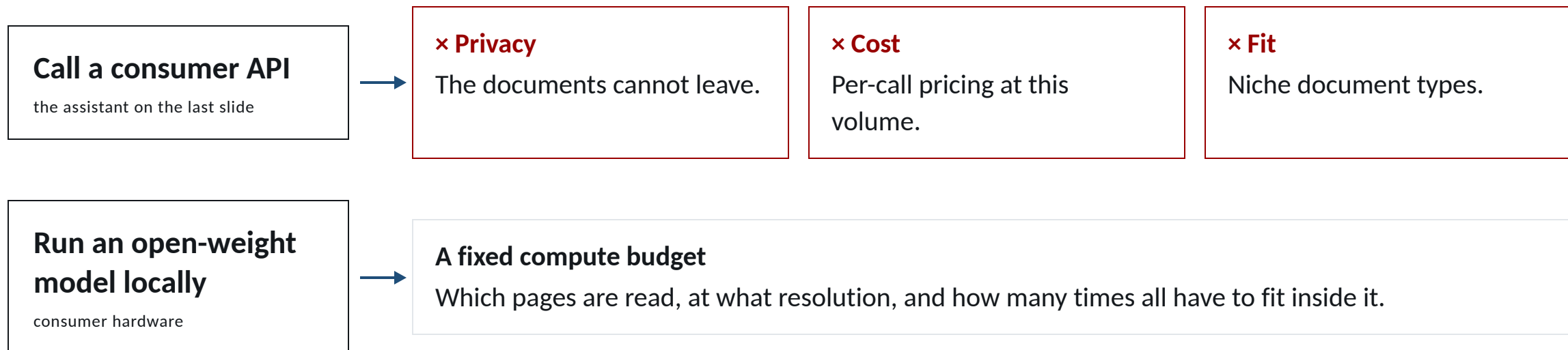
Completed task **Verify and answer**

- A question over a **100-page thesis**: the assistant plans, locates the pages, then answers.
- It **renders the pages and reads them as images**.



Real-World Constraints

A hospital, a mining company, a law firm: **tens of thousands of documents** to classify, summarise, or extract from.



The research question: document understanding **under a fixed compute budget**.

Contributions

A thesis by compilation, in three parts.

CONTRIBUTION 1

A survey

A taxonomy of the field through **evidence management**: four architectural families, and the strategies built around them.

CONTRIBUTION 2

An empirical attribution

A framework of **three failure loci**, each tested by changing one thing in one pipeline.

CONTRIBUTION 3

A proof of concept

A training-free method built from the findings: **56.5%** on MMLongBench-Doc, the highest on its backbone.

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- Introduction

- **Survey of MP-VRDU**

Managing Evidence at Document Scale: A Survey of Multi-Page Visually Rich Document Understanding.

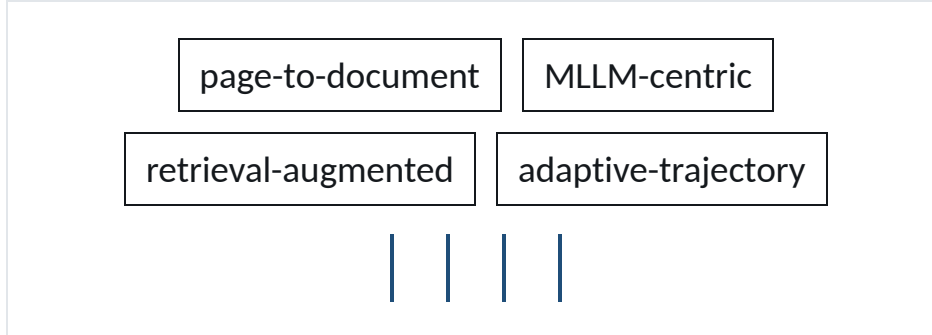
Lewei Xu, Yihao Ding, Zihan Xu, Siwen Luo, Daochang Liu, Wei Liu

EMNLP 2026, main conference

- Empirical Attribution Study
- Proof of Concept: A Simple Method
- Conclusion

Survey: Research Gap and Contributions

RESEARCH GAP



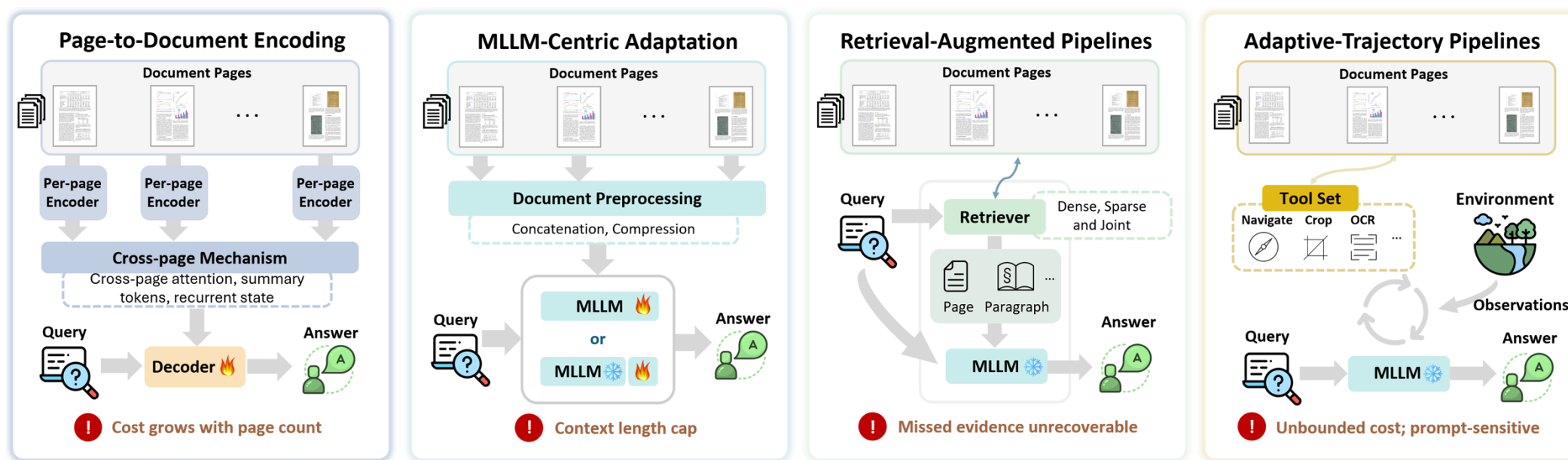
Multi-page systems grew along **parallel lines**, with inconsistent terminology. No survey treats multi-page as its own problem.

CONTRIBUTIONS

- 1 Defines multi-page understanding as a problem of **evidence management**.
- 2 A taxonomy: four architectural families, multimodal representation, inference-time and training strategies.
- 3 Consolidates the datasets and five open challenges.

Multi-Page Architectural Overview

The survey organises systems by **evidence management**, into four architectural families. **Page-to-document encoding models** jointly encode pages; **MLLM-centric adaptation models** fit multi-page input into an MLLM's context and token budget; **retrieval-augmented pipelines** retrieve evidence before generation; **adaptive-trajectory pipelines** inspect evidence iteratively under an LLM controller.



Snowflake indicates typically frozen, fire indicates typically trained

ROUGHLY CHRONOLOGICAL 2023 Page-to-document encoding ► 2024 MLLM-centric, retrieval-augmented ► 2025 Adaptive-trajectory

Inference-Time and Training Strategies

Three evidence-management decisions. Each can be **prompted** around a frozen MLLM, or **trained** into it.

Retrieval and navigation

TRAINING-FREE

Similarity-based or relation-aware retrieval

TRAINED

Modality-alignment, cross-page-fusion, logical-relevance

Reasoning strategies

TRAINING-FREE

Chain-of-Thought, Self-Reflection, Self-Consistency

TRAINED

Trace-imitation, then reward-driven (GRPO)

Agentic methods

TRAINING-FREE

ReAct and tool use, multi-agent orchestration

TRAINED

Trajectory-distillation, reinforcement-learned

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- **Empirical Attribution Study**

Locating Failure in Multi-Page Visually Rich Document Understanding: An Empirical Attribution

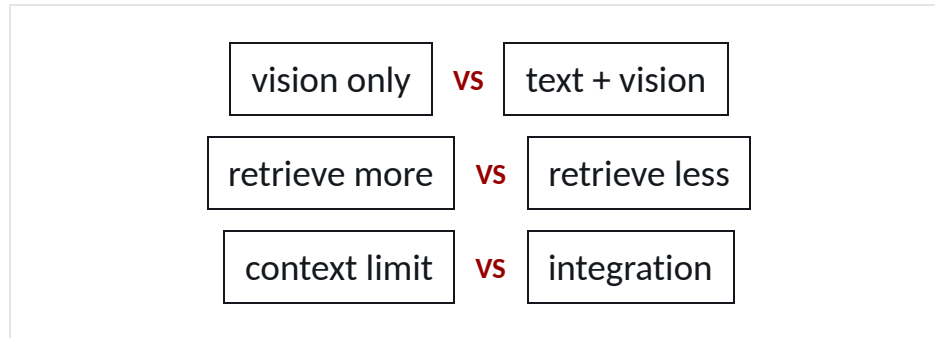
Lewei Xu, Yihao Ding, Zihan Xu, Daniel Yitian Su, Daochang Liu, Siwen Luo, Yifan Peng, Wei Liu

Submitted to EACL, under review

- Proof of Concept: A Simple Method
- Conclusion

Empirical Study: Research Gap and Contributions

RESEARCH GAP



Papers make **competing claims** about what matters, each on a different pipeline and dataset, so none can be compared.

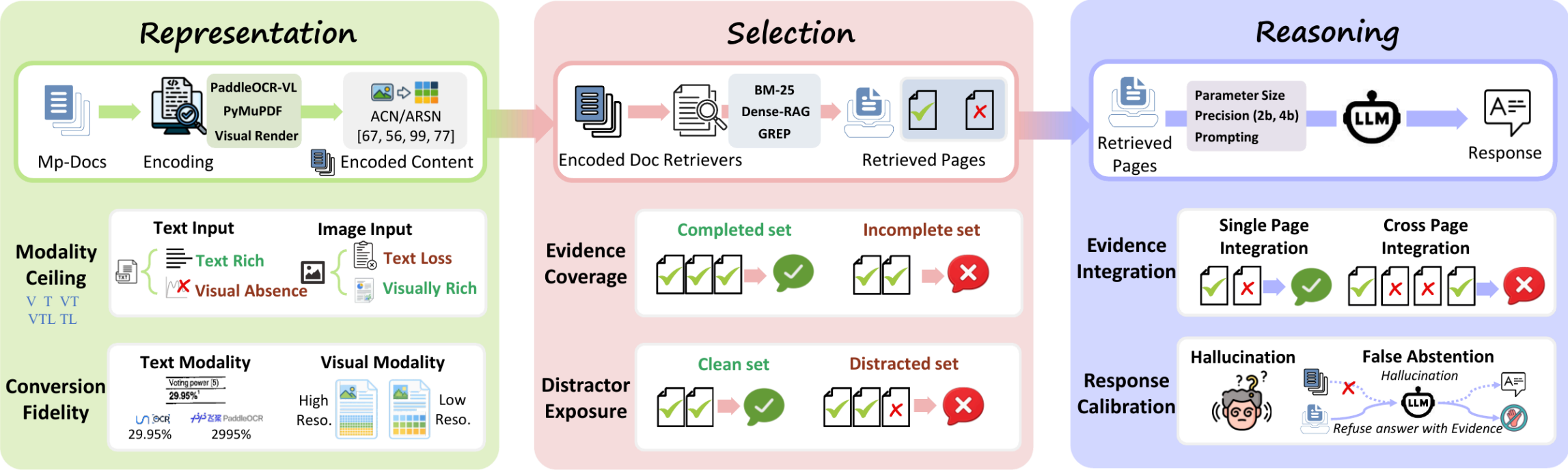
CONTRIBUTIONS

- 1 Three architecture-independent **failure loci**, with six mechanisms.
- 2 Controlled interventions in one pipeline: change one locus, hold the others fixed.
- 3 Transfer across datasets and models, with design implications and cost.

Correct answer: **6**. Suppose the system says **4**.

End-to-end accuracy **cannot tell these apart**, and each needs a different fix.

Attribution Framework



REPRESENTATION

Was the information preserved when the page was converted?

SELECTION

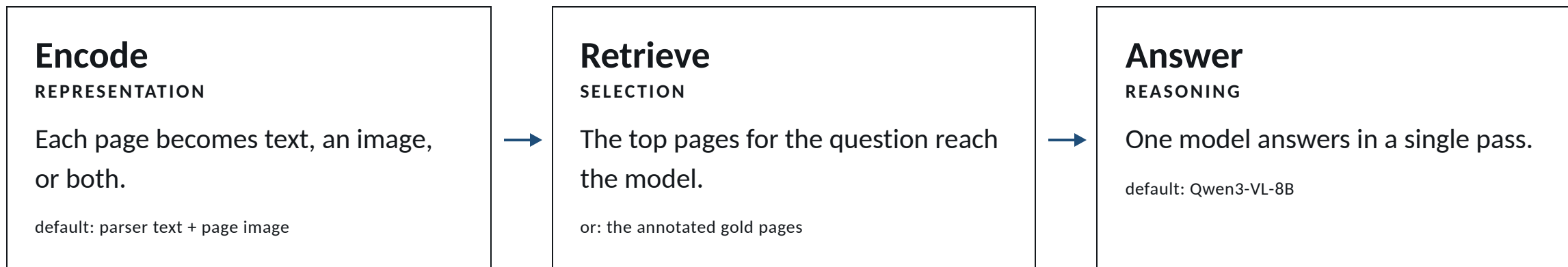
Did it actually reach the model?

REASONING

Did the model use it correctly?

Attribution by Construction

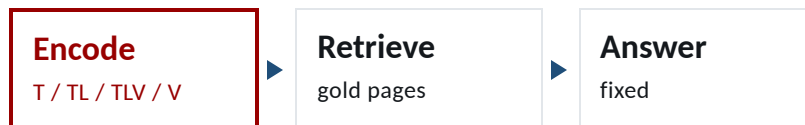
One simple pipeline. **Change one stage, hold the other two fixed.**



- **Gold pages** bypass retrieval, so selection can be taken out of the picture.
- **MMLongBench-Doc**: 135 documents, annotated with evidence pages, evidence type and answerability.

Modality Ceiling

Evidence	Abstention				Accuracy			
	T	TL	TLV	V	T	TL	TLV	V
Text	23.0	11.5	7.3	8.0	36.9	44.4	51.0	42.3
Layout	20.3	10.2	0.8	4.2	26.3	33.9	49.2	43.2
Table	15.9	7.2	6.7	9.1	35.6	44.7	44.7	34.1
Chart	32.6	28.1	7.3	6.2	16.3	16.9	42.7	40.4
Figure	36.6	24.8	7.6	7.9	14.5	22.1	42.4	39.0
<i>All</i>	26.3	17.2	6.2	7.1	29.0	35.7	49.5	42.6



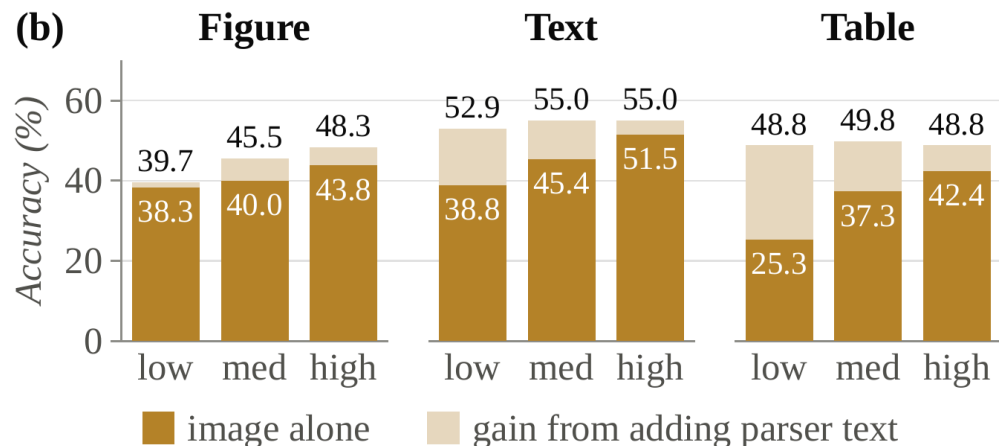
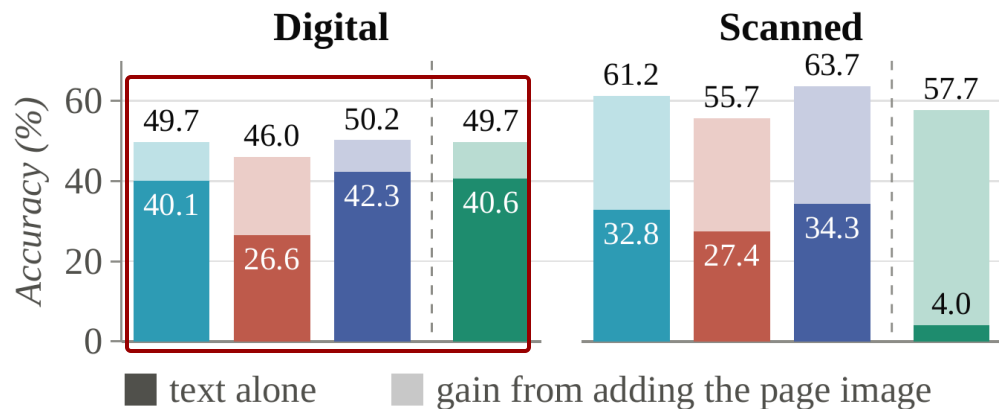
REPRESENTATION

Evidence accessibility depends on modality

- Text-only input refuses **chart and figure** questions most. Adding vision brings refusal **under 10%**.
- Images alone stay below text plus images: **42.6% against 49.5%**.

Conversion Fidelity

(a) PaddleOCR MinerU Unlimited-OCR PyMuPDF



Encode
parser, resolution

Retrieve
gold pages

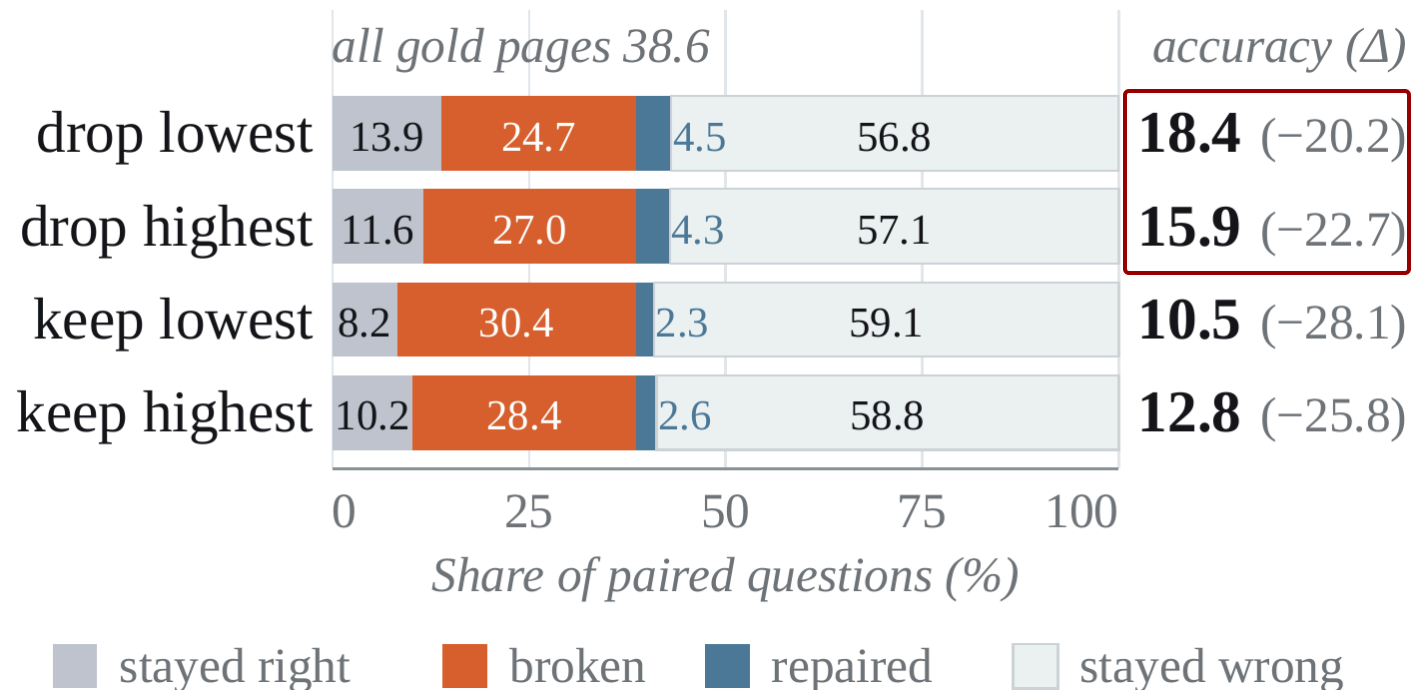
Answer
fixed

REPRESENTATION

Complementary modalities reduce conversion sensitivity

- The parser spread on digital documents shrinks from **15.7 to 4.2 points** once page images are added.
- Higher resolution helps images alone far more than text plus images.

Evidence Coverage

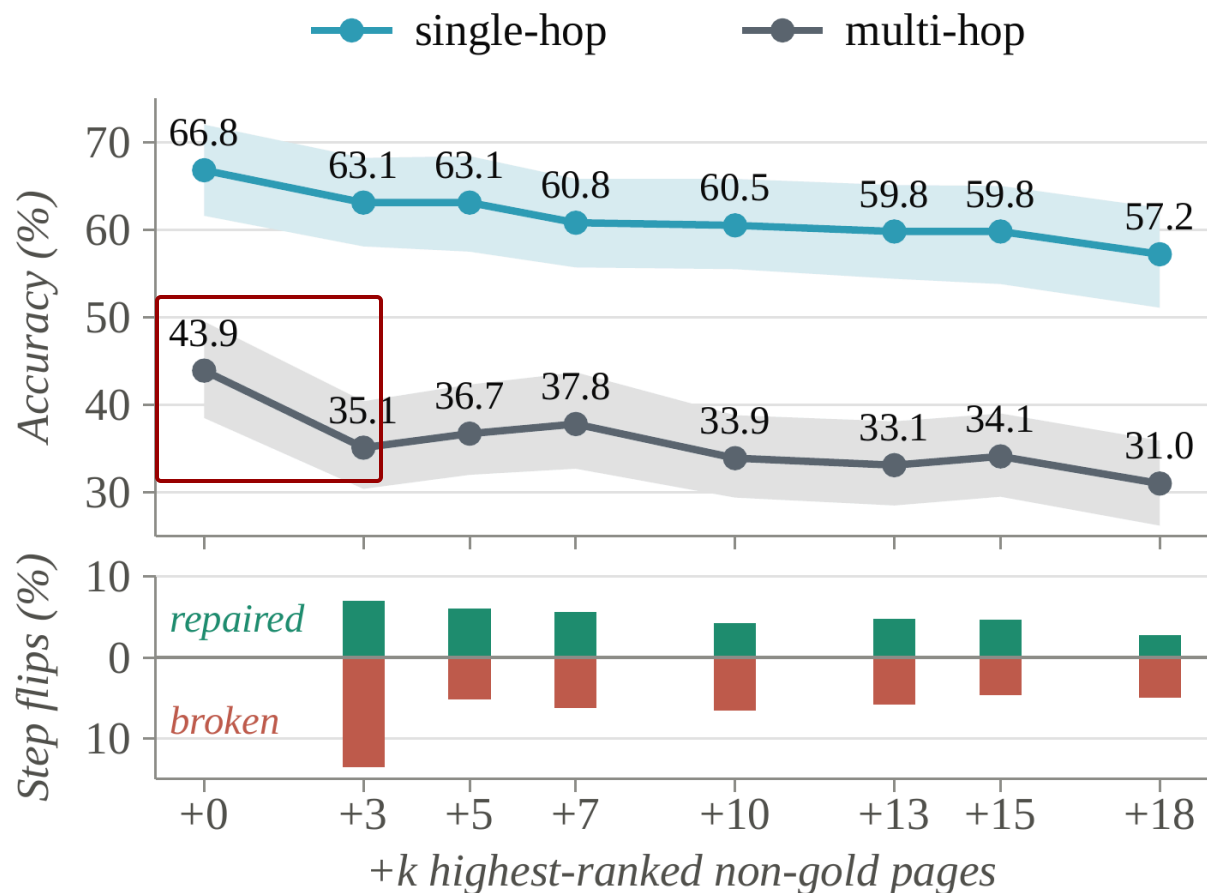


SELECTION

Omission causes large accuracy losses

- Removing **one gold page** takes accuracy from **38.6% to 18.4%** or 15.9%.
- Retrieval rank does not predict which page is safe to lose.

Distractor Exposure



Encode
fixed

Retrieve
gold + k other pages

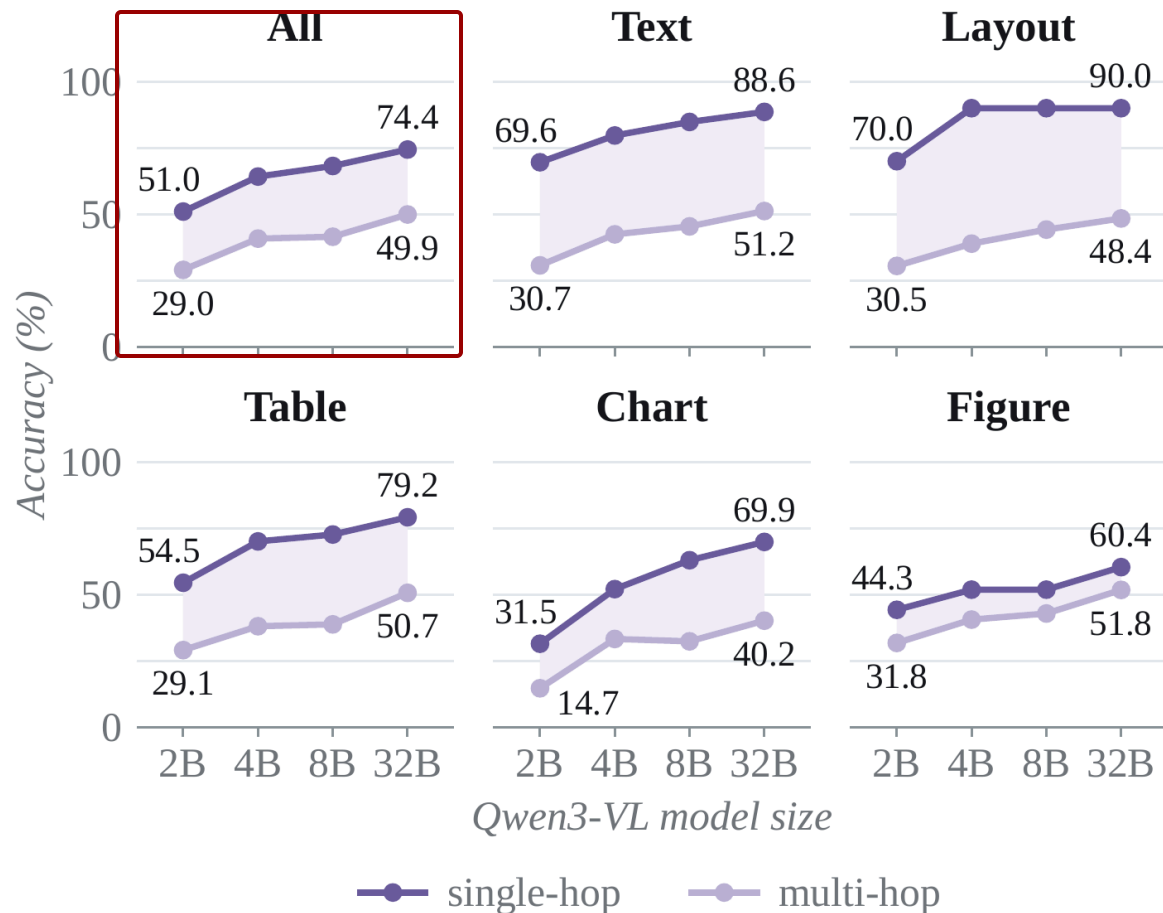
Answer
fixed

SELECTION

Multi-hop accuracy drops sharply with added context

- Three extra pages: multi-hop falls from **43.9%** to **35.1%**. Single-hop declines gradually.
- Every gold page is still there. Added pages both break and repair answers.

Evidence Integration

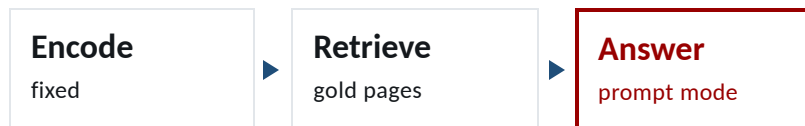
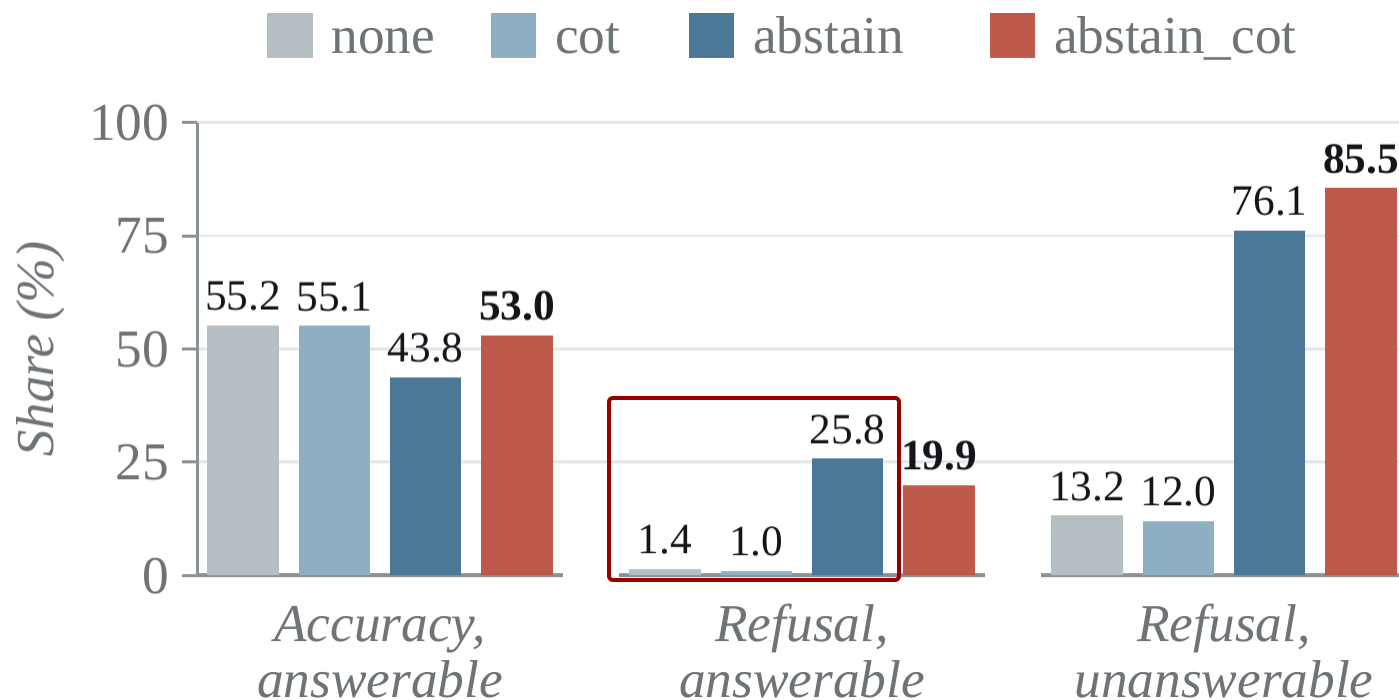


REASONING

Scale improves multi-hop performance

- From 2B to 32B, multi-hop accuracy rises from **29.0% to 49.9%**.
- Single-hop rises too, so scale helps evidence use in general.

Response Calibration



REASONING

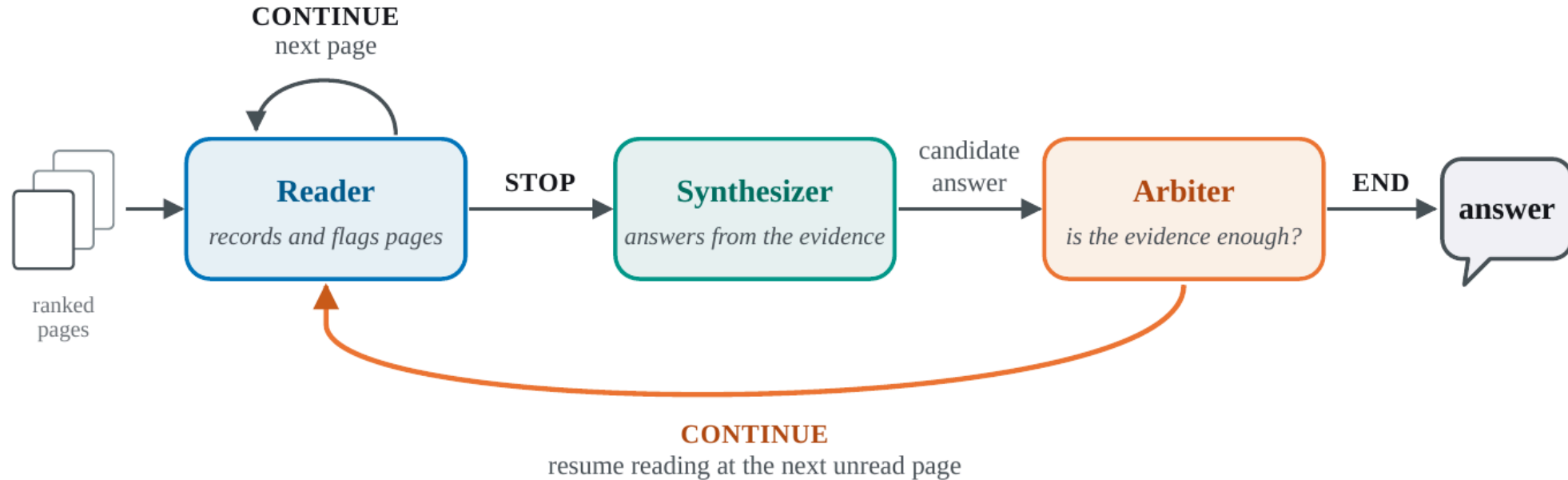
Reasoning before abstaining limits false refusal

- An abstain instruction lifts correct refusal from 13.2% to 76.1%, but false refusal goes from **1.4% to 25.8%**.
- Adding CoT recovers answerable accuracy to 53.0%.

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From Findings to a Simple Method



REPRESENTATION

Every page is read as **text and image together**.

SELECTION

Read broadly, then pass on compact records, not full pages.

REASONING

A separate arbiter decides **whether the evidence is enough**.

Simple Method Results

Method	OCR	Trained	MMLongBench-Doc				LongDocURL			
			SIN	MUL	UNA	ACC	UND	REA	LOC	ACC
Gold pages + VLM	✓	×	57.7	37.4	52.5	49.5	69.7	63.6	59.5	65.7
Whole document (via DocTrace)	×	×	46.9	35.3	25.8	38.5	—	—	—	45.1
M3DocRAG (via SLEUTH)	×	×	—	—	48.6	45.9 ^a	60.1	52.4	45.5	54.6
DocAgent (via MARDoc)	✓	×	—	—	—	46.6	—	—	—	—
MDocAgent (via SLEUTH)	✓	×	—	—	49.7	47.8 ^a	57.5	51.5	44.7	53.1
MoLoRAG (via SLEUTH)	×	×	—	—	51.7	48.8 ^a	63.4	52.7	49.3	57.6
SLEUTH	×	×	—	—	67.4	52.8 ^a	65.7	53.0	53.6	60.0
DocTrace	✓	✓	53.2	41.1	70.5	52.9	—	—	—	56.4
Ours (4-bit)	✓	×	60.5	39.2	76.9	56.5	67.2	66.3	47.9	61.3

- **56.5%** on MMLongBench-Doc and **61.3%** on LongDocURL: highest among the Qwen3-VL-8B systems, training-free, at 4-bit.
- Multi-page questions still trail DocTrace, **39.2 against 41.1**.

Conclusion

SURVEY

Organise

Multi-page document understanding is a problem of **evidence management**.

EMPIRICAL STUDY

Locate

Failures sit at **representation, selection, or reasoning**, and each needs a different fix.

PROOF OF CONCEPT

Validate

The findings hold up in a **working, training-free system**.

Thank you

Questions?

Retrieval and Navigation

Retrieval **narrows a long document to compact evidence units** (chunks, pages, or graph nodes) for downstream reasoning.

- **Similarity-based retrieval.** Scores each evidence unit independently against the query and returns the top-ranked subset. Sparse lexical signals such as BM25 are the baseline; dense retrieval uses vision-language late-interaction encoders such as ColPali and ColQwen, often with an LLM reranking stage after.
- **Relation-aware retrieval.** Retrieves along explicit document structure. Tree-based variants follow section hierarchies and outlines, so chunks keep their parent context; graph-based variants traverse typed edges between headings, entities, and cross-references.
- **The trade-off.** Independent scoring handles paraphrase and cross-modal queries, but cannot follow cross-page references or multi-hop dependencies. Relation-aware methods can, at the cost of parser dependence and traversal cost.

Evidence missed or poorly ranked at retrieval is **unrecoverable at generation**. What is retrieved still has to be reasoned over.

Reasoning Strategies

Reasoning strategies act over a **single evidence buffer assembled before the reasoning call**: the full document representation, a retrieved subset, or evidence from an agentic step. None updates model parameters.

- **Sequential reasoning: Chain-of-Thought.** Extends a single reasoning thread. CoT binds evidence from different pages into one chain and decomposes multi-hop questions into single-hop steps.
- **Sequential reasoning: Self-Reflection.** The model evaluates its own draft answer and revises it across rounds, optionally against a verifier or critic.
- **Parallel exploration.** Several reasoning paths over the same buffer. Self-Consistency samples multiple chains and takes the majority answer; sampling-adjudication replaces the vote with an LLM judge. Cost scales with the number of samples. Tree-of-Thoughts has seen little uptake.

A chain can only reason over the evidence already assembled. It **cannot recover what retrieval missed**.

Agentic Methods

Agentic methods put one or more **LLM controllers** in charge: either iterating evidence acquisition under inference-time control, or coordinating across specialised roles.

- **ReAct and tool-augmented reasoning.** A single controller interleaves thought, action, and observation. Actions are document-side tool calls (retrievers, parsers, OCR engines) or structured navigation over a pre-built document representation.
- **Multi-agent orchestration.** Modality-specialised decomposition gives text and visual evidence separate controllers plus a synthesis controller. Role-specialised decomposition splits planning, execution, verification, and synthesis.
- **What the adaptive trajectory buys.** Compute scales with question difficulty rather than document length, the single-step recall ceiling of retrieval-augmented pipelines is lifted, and the action sequence is an inspectable audit trail.

The cost is **controllability**: inference cost is hard to bound, and a controller can commit early to wrong evidence and continue down the wrong trajectory.

Training Strategies

Most systems adopt a backbone **already pretrained on single-page data**, so fine-tuning carries the weight. Training internalises the inference-time evidence-management decisions into model parameters.

Trained retrieval

Modality-alignment aligns visual page embeddings with OCR-derived text. **Cross-page-fusion** adds neighbouring-page context to each page representation. **Logical-relevance** ranks pages by their position in a cross-page reasoning path.

RETRIEVER

Trained reasoning

Trace-imitation supervises the backbone on demonstration chains in a fixed schema. **Reward-driven** training then refines it with GRPO, rewarding answer accuracy, evidence-page recall, and format compliance.

REASONER

Trained agentic methods

Trajectory-distillation clones a stronger teacher's think-action-observation turns through SFT. **Reinforcement-learned** variants optimise the policy directly against trajectory-level rewards.

CONTROLLER

Two techniques recur across all three: **parameter-efficient tuning** (LoRA, adapters) and **multi-stage curricula** that go from single-page to document-scale supervision. All of it depends on the available datasets.

Datasets

Datasets serve three roles, and the multi-page setting makes all three harder: **per-page annotation does not scale** to documents of tens or hundreds of pages, and long PDFs are harder to source and license than single pages.

Pretraining

The exception rather than the norm. Confined to a few page-to-document and MLLM-centric systems that do continued pretraining on an existing backbone, because multi-page corpora are still small.

RARE

Fine-tuning

Has converged on LVLM-synthesised supervision over scraped PDFs, with reinforcement learning increasingly layered on top to shape evidence-grounding and trajectory behaviour.

LVLM-SYNTHESISED

Benchmarking

Three tiers: single-page extractive baselines, mid-length multi-page suites, and long-document multimodal benchmarks that need cross-page integration. The same small pool of PDFs gets reused.

CONTAMINATION RISK

Answer-only accuracy masks retrieval failure and rewards correct answers reached through incorrect reasoning. Newer benchmarks score retrieval and page-localisation as well, but it is not yet standard.

Experimental Setup

One simple pipeline throughout, with **one condition changed at a time** and everything else held fixed.

Encode

Every page becomes text, an image, or both.
Done once per document and cached, so no question changes what a page contains.

PaddleOCR-VL · 200 DPI

Retrieve

▶ A retriever scores every encoded page against the question and returns the top k , restored to document order.

ColQwen3 · page-level

Answer

▶ A multimodal model receives those pages and the question, and answers in a single pass.
Nothing is fed back.

Qwen3-VL-8B · greedy

- **The stages act once, and in order.** That is what lets a failure be attributed rather than guessed at.
- **Gold pages bypass retrieval.** For most tests the annotated evidence pages are supplied directly, so selection is neutralised and whatever remains is representation or reasoning.
- **Answers are scored by an LLM judge**, Gemini-2.5-Flash at temperature zero. It agrees with GPT-4o on 96.0% and with a human annotator on 98.3% of validation subsets.

Datasets and Evaluation

None of these experiments are possible without annotation that says **which pages support each question**. That is what picks the benchmarks.

MMLongBench-Doc

PRIMARY DIAGNOSTIC

135	47.5	847	244
documents	mean pages	answerable	unanswerable

The only multi-page benchmark we know of that annotates, per question, **the evidence pages, the kind of evidence it draws on, and the document's domain**, while also supplying plausible questions the document cannot answer.

Evidence pages make coverage measurable and define the oracle. Evidence source explains representation results by what the question actually needs. The unanswerable pool makes calibration testable rather than inferred.

Answers are scored by an LLM judge rather than exact match, since a correct response can differ in units, rounding, or formatting. The judge sees only the question, gold answer, and model answer, never the condition being tested. It agrees with GPT-4o on **96.0%** of a stratified sample and with a human annotator on **98.3%**. We track refusals and wrong answers separately, not just accuracy.

LongDocURL

SECONDARY REPLICATION

396	85.6	2,325	8
documents	mean pages	questions	unanswerable

Longer documents, used to check whether the findings hold under a **less demanding setting**. Born-digital throughout, almost no unanswerable questions, and many questions answerable from the text even when the annotated evidence is visual.

Its annotated pages are also not a minimal set: several may be redundant or independently sufficient, which we account for when reading its coverage and integration results.

LongDocURL Transfer

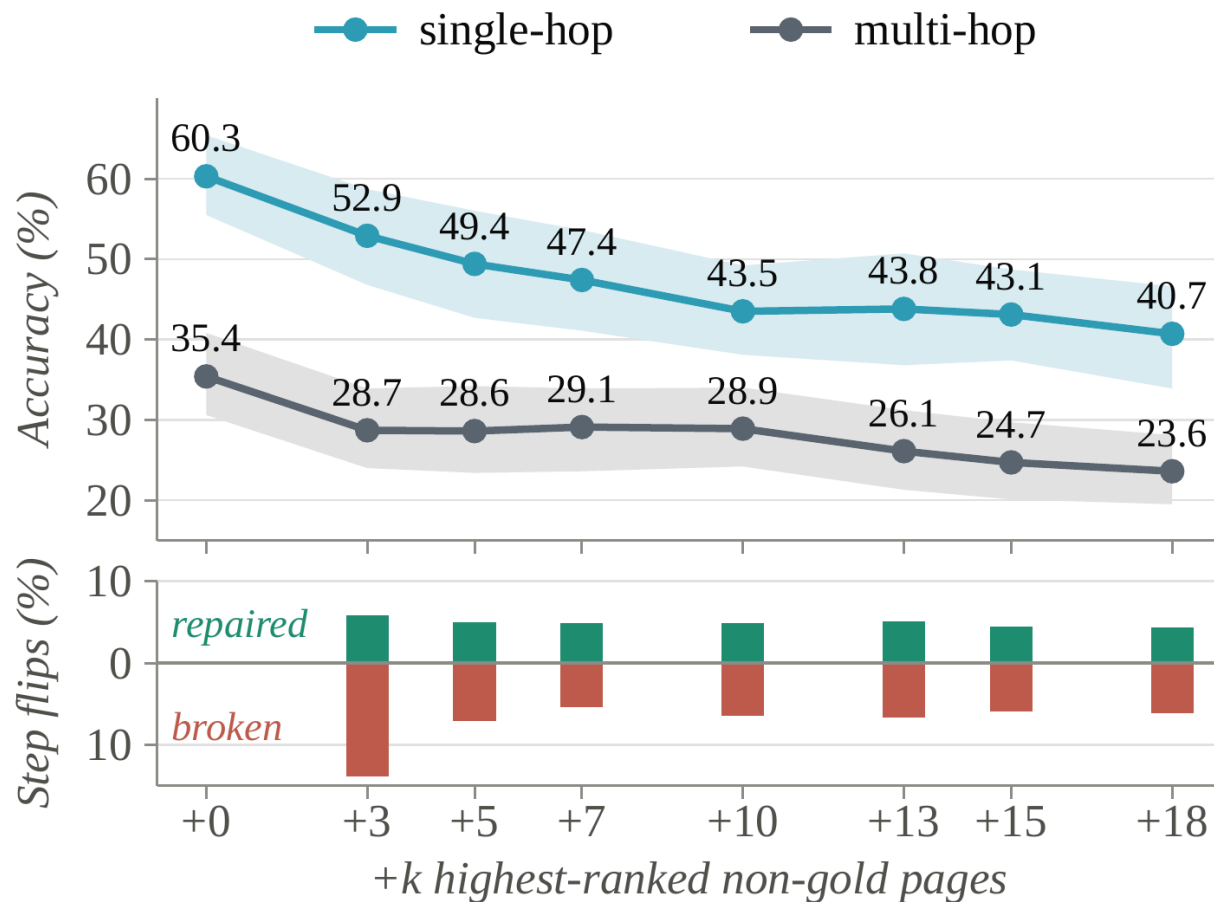
	T	TL	TLV	V
<i>LongDocURL</i>				
Text	75.3	79.5	82.3	74.0
Layout	48.7	52.4	56.3	52.2
Table	61.2	69.5	68.2	57.3
Figure	62.5	65.7	73.4	65.2
All	63.7	68.0	71.9	63.1
<i>MMLongBench-Doc</i>				
Born-digital	40.6	40.1	49.7	41.5
Scanned	4.0	32.8	61.2	58.2

REPRESENTATION

Born-digital documents leave less for vision to add

- Figure questions already reach **65.7%** from parser text. Adding images gives **3.9 points** overall.
- On MMLongBench-Doc, vision adds 28.4 points on scans and 9.6 on born-digital documents.

Distractors on Another Backbone

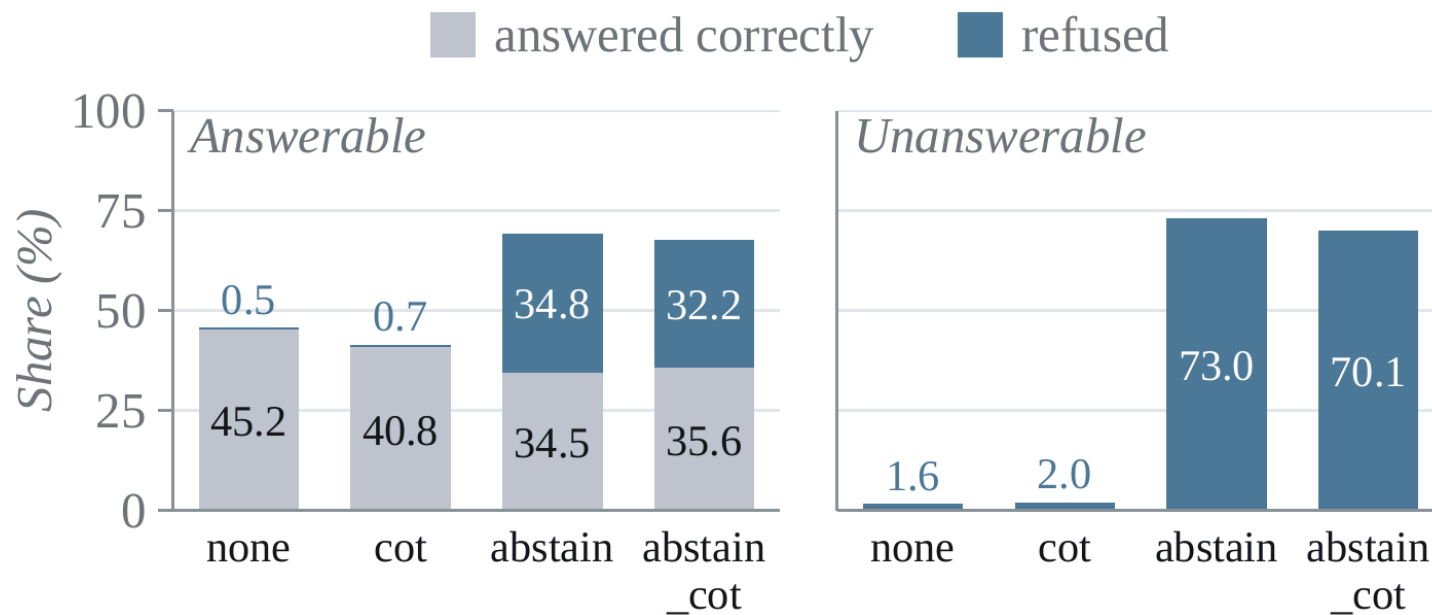


SELECTION

Gemma3-12B shows the same decline

- Single-hop falls from **60.3%** to **40.7%**, multi-hop from **35.4%** to **23.6%**.
- Added context still both repairs and breaks answers.

Calibration on Another Backbone



REASONING

Prompt effects depend on the model

- Adding CoT to abstain gives Gemma3-12B **1.1 points** of answerable accuracy, against 9.2 for Qwen3-VL-8B.
- Correct refusal falls 2.9 points for Gemma and rises 9.4 for Qwen.

Representation Cost

	Acc.	Tokens per page			Latency (s)	
		Text	Vis.	Total	Prefill	Decode
T	29.0	499	0	499	1.4	5.7
TL	35.7	874	0	874	1.8	6.6
TV	49.6	500	459	959	—	—
TLV	49.5	874	459	1333	26.3	6.7
V	42.6	20	459	479	29.0	7.1

DEPLOYMENT

Latency follows visual processing, not token count

- Prefill is **29.0 s** per question with images alone, against 1.4 to 1.8 s for text.
- Embedded text plus images matches parser text plus images: **50.1% against 49.5%**, with no parser.

Reasoner Choice Under Budget

Reasoner	T	TL	TLV	V	GB
<i>Scale (Qwen3-VL, bf16)</i>					
2B	24.7	31.9	38.3	31.9	4.3
4B	32.3	38.6	50.4	42.4	8.9
8B	31.9	38.4	52.4	45.5	17.5
32B	35.5	43.9	59.6	56.7	66.7
<i>Quantized (Qwen3-VL)</i>					
8B, 8-bit	32.5	37.9	53.6	45.7	10.3~
8B, 4-bit	31.8	39.4	51.8	45.5	6.8~
32B, 4-bit	37.0	42.1	59.1	52.3	20.0~
<i>Other families (bf16)</i>					
InternVL3-8B	19.3	25.3	32.6	24.1	15.9
MiniCPM-V-4.5 (8B)	30.7	38.1	60.1	53.3	17.4
GLM-4.6V-Flash (9B)	25.9	32.9	42.7	30.9	20.6
Llama-3.2-11B-Vision	26.5	34.3	33.3	25.3	21.3
Gemma-3-12B	30.5	36.9	48.1	41.6	24.4

DEPLOYMENT

Family and precision matter as well as size

- Qwen3-VL-8B keeps **51.8%** at 4-bit, in about 6.8 GB of weights.
- The 4-bit 32B model reaches **59.1%** in about 20 GB.

From Findings to Design

The survey and the experiments both point at design principles. I put them together in a **deliberately simple system**, to see whether they hold up at the level of a whole pipeline rather than one controlled condition.

REPRESENTATION

Keep text and images together

- ▶ Every page read at the full TLV representation, with a strong parser and high-resolution rendering, since the two channels cover for each other's failures.

SELECTION

Gather broadly, present selectively

- ▶ Missing evidence is unrecoverable and moderate distraction is not, so read widely in retrieval-ranked order, but carry most of it forward as compact records rather than as full pages.

REASONING

Check whether the answer is supported

- ▶ Because the reasoner cannot judge sufficiency on its own, a separate step decides whether to accept the answer or keep reading.

No planner, no tool set, no query reformulation. **The point is to validate the principles, not to introduce a complex architecture.**

Simple Method Cost

	Turns	Tokens			Latency (s)		Peak VRAM (GiB)
		Input text	Input visual	Output	Prefill	Decode	
reader	11.5	5,030	816	697	1.8	46.8	7.2
synthesizer	1.2	9,307	2,018	463	3.8	34.7	8.1
arbiter	1.2	6,806	1,734	464	2.9	33.3	7.9
Per question, mean	13.8	76,669	13,767	9,092	28.5	617.5	8.2
Per question, median	9.0	43,113	9,830	6,078	15.8	385.0	8.0

- **13.8 turns** per question on average, 11.5 of them the reader.
- The cost is time, not memory: **10.8 minutes** mean, 6.7 median, with peak memory at 8.2 GiB.